

GEOGRAPHY

PHYSICAL GEOGRAPHY



CLIMATOLOGY

MODULE – 11

TEMPERATURE
FACTORS CONTROLLING
TEMPERATURE (PART-1)

CLIMATOLOGY MODULES

Atmosphere	▪ Composition and structure; weather and climate;
Insolation	▪ Factors controlling insolation; heat budget of the earth — heating and cooling of atmosphere (conduction, convection, terrestrial radiation, advection);
Temperature	▪ Factors controlling temperature; distribution of temperature — horizontal and vertical; inversion of temperature;
Pressure	▪ Pressure belts; winds-planetary, seasonal and local; air masses and fronts; tropical and extra-tropical cyclones;
Precipitation	▪ Evaporation; condensation — dew, frost, fog, mist and cloud; rainfall — types and world distribution;
World Climate and Climatic Changes	▪ Different types of Climate



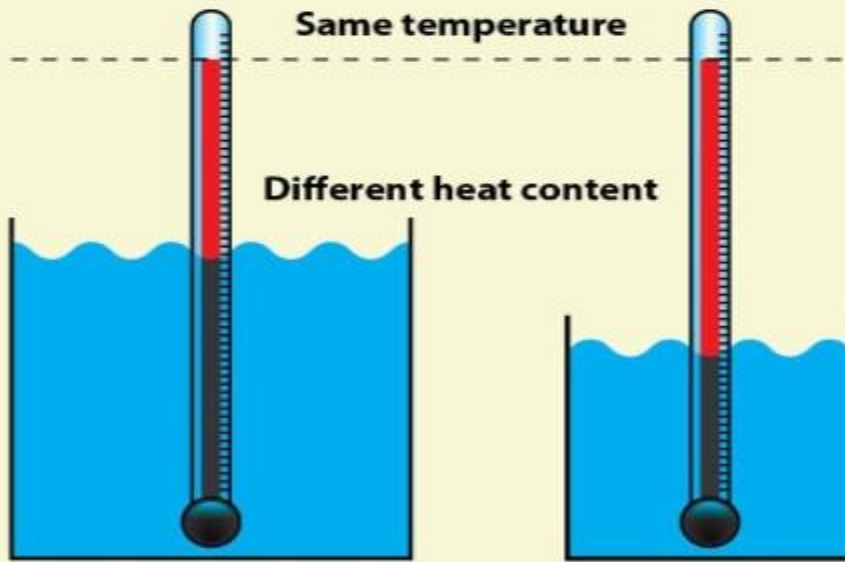
WHAT IS TEMPERATURE?



HEAT



TEMPERATURE



- 1 Heat $\neq$ Temperature; appear synonymous but in reality very different from one another.
- 2 Heat = quantity of energy present in any substance
- 3 Temperature = degree of hotness or coldness
- 4 The interaction of insolation with the atmosphere and the earth's surface creates heat which is measured in terms of Temperature.
- 5 In fact, the atmosphere receives very low amount of heat energy from the sun as it receives most of its energy from the long-wave terrestrial radiation.



WHAT IS TEMPERATURE?



IMPORTANT CONCEPT FOR UPSC PRELIMS MCQs

1. Factors controlling **DISTRIBUTION OF INSOLATION** (done in Module 9)
2. Factors controlling **DISTRIBUTION OF TEMPERATURE**
3. Explicit MCQs on **DIFFERENTIAL RATE OF COOLING AND HEATING** in
land and water



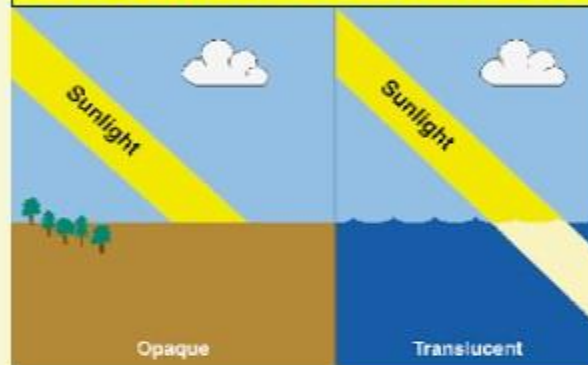
FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



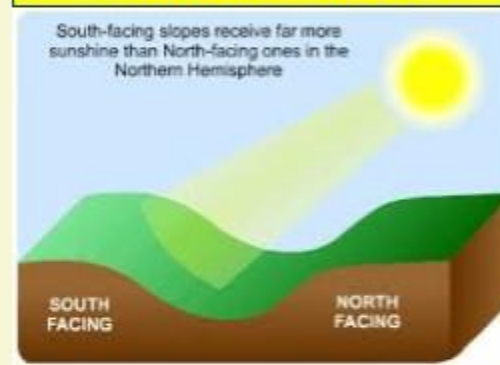
1. LATITUDES



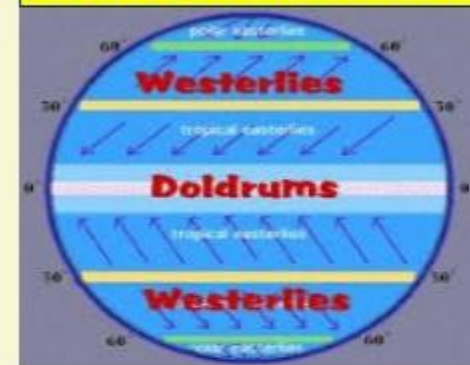
3. NATURE OF LAND & WATER



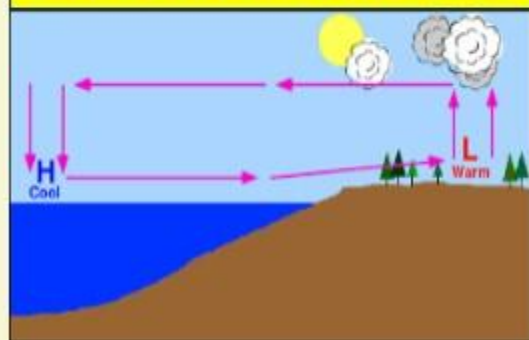
5. GROUND SLOPE



7. PREVAILING WINDS



2. DISTANCE FROM SEA



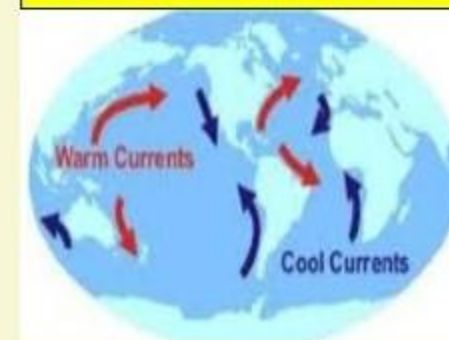
4. ALTITUDE



6. GROUND SURFACE



8. OCEAN CURRENTS



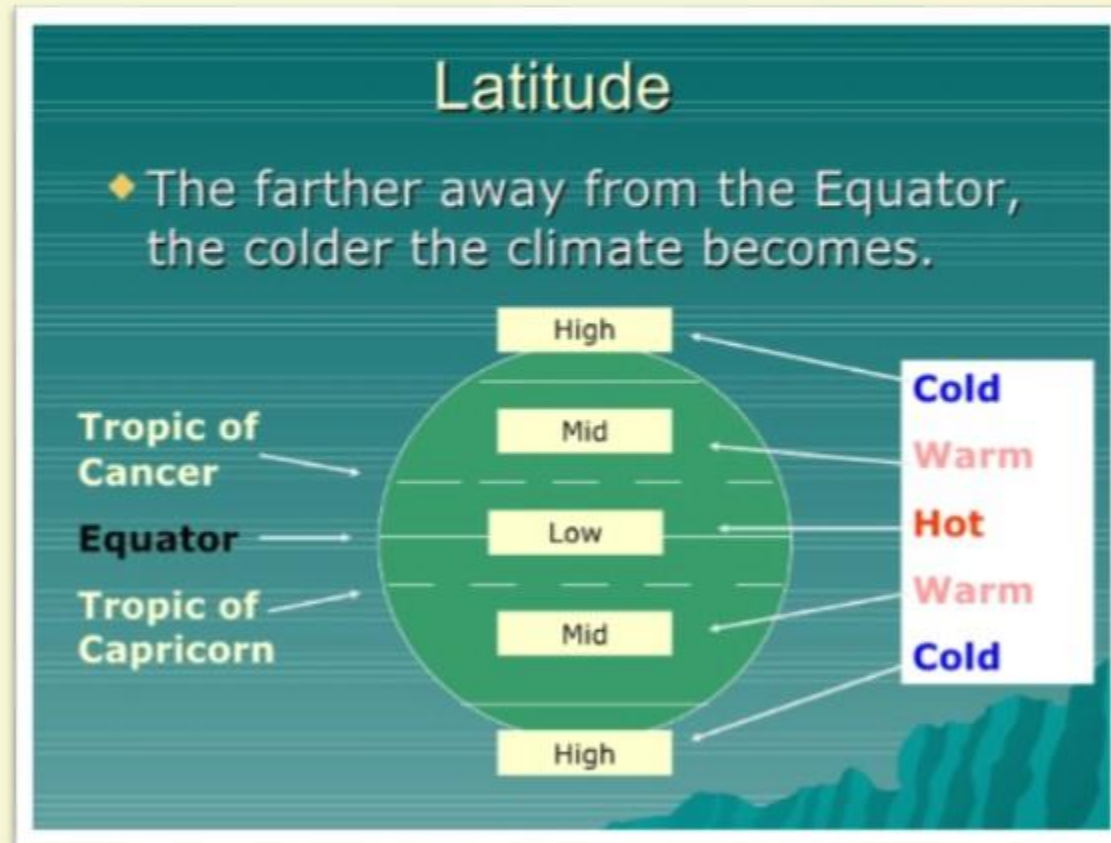


FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 1

LATITUDES OF THE PLACE



- 1 The temperature of a place depends on the insolation received in a place.
- 2 Since the amount of insolation received in a place decreases from tropics (direct rays) towards the poles (oblique rays) and hence air temperature also decreases poleward. Thus it is said that temperature depends upon the latitude of the place.



FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 1

LATITUDES OF THE PLACE

So by this analogy, shouldn't the equator have the highest temperature?



- Actually NO.
- The maximum temperature is not recorded on the equator but rather on 20°N.

Why is it so? (Probable UPSC question)

1. Sizeable amount of heat lost in evaporation
2. Cloud cover reflects some part of the insolation
3. Vapour condenses in the later part of the day to form thick cumulonimbus clouds(rain bearing clouds) and leads to heavy rainfall.

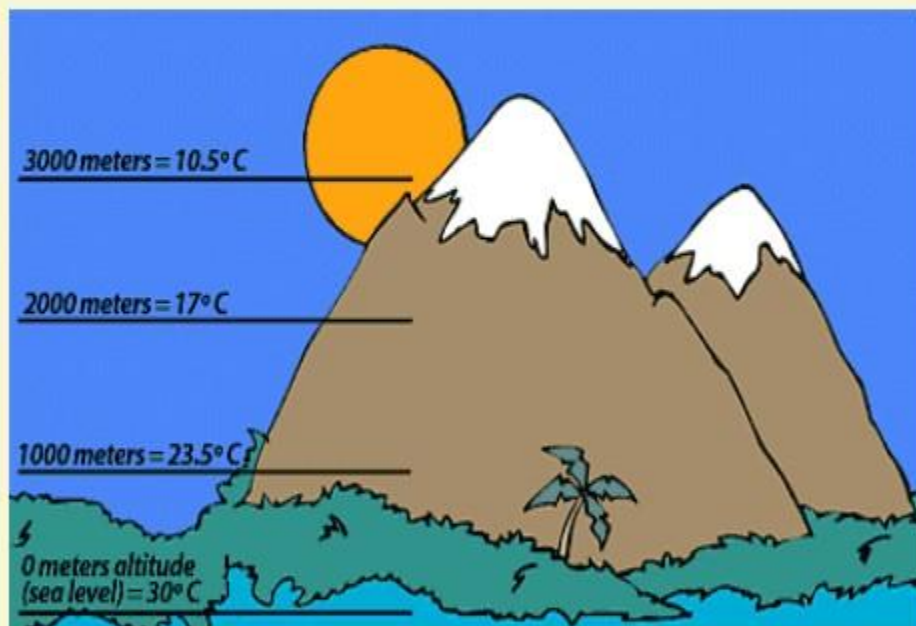


FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 2

ALTITUDE OF THE PLACE



- 1 Temperature decreases as height increases at 6.5°C per km(NLR). This is because of the following reasons:

- 2 As we ascend higher in the atmosphere, the amount of heat gets lost to successive layer lying below.
- 3 The air pressure is higher in the lower portion of the atmosphere near the earth's surface because of the weight of all the overlying layers above. Thus density of air decreases as we move upwards.
- 4 The lower layer of the air contains more water vapour and dust particles than the layers above and hence it absorbs larger amount of heat radiated from the earth's surface than the upper layers.



FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 2

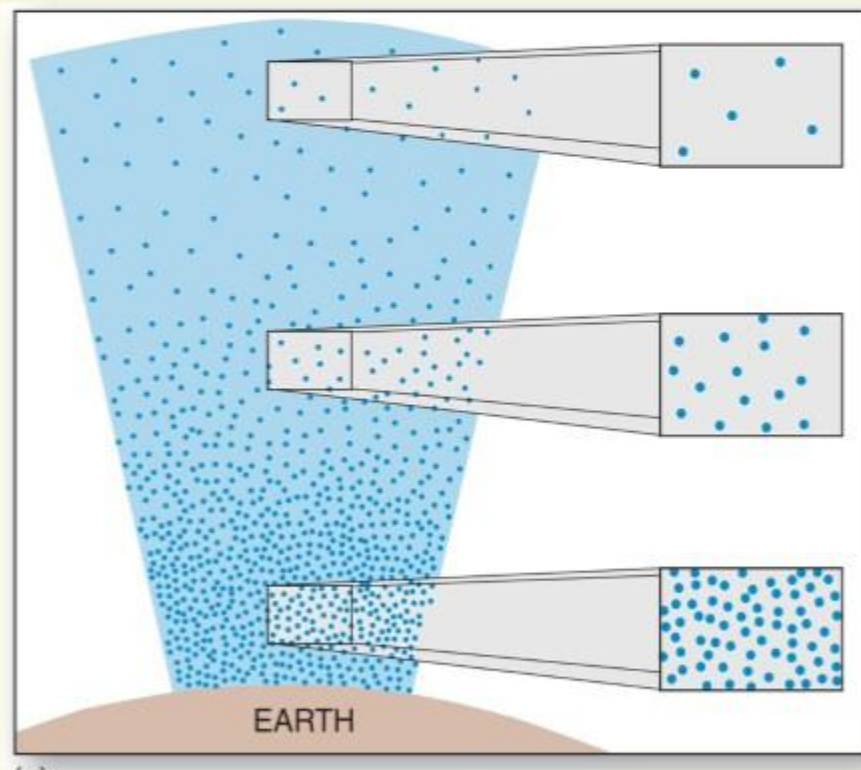
ALTITUDE OF THE PLACE

*Connection between Altitude,
Temperature and Density
in the Air above us.*

*Altitude
Increases*



*Air Temperature
& Density Decreases*



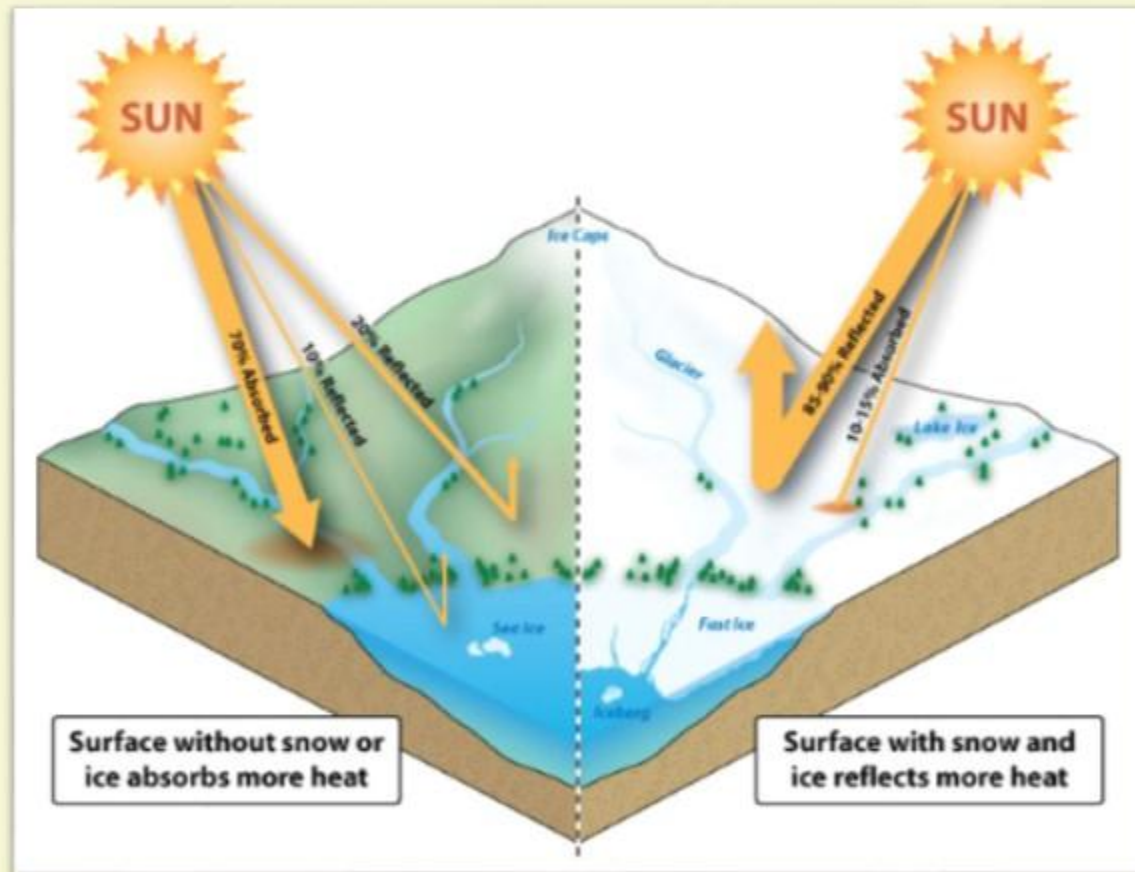


FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 3

NATURE OF GROUND SURFACE



- 1 The nature of ground surface in terms of **colour, vegetation and land use practices** etc affect the distribution of temperature.
- 2 The **snow-covered surface** receive **very low amount of insolation** because they reflect 70-90% of incoming shortwave radiation.
- 3 **Thick vegetative foliage** cuts off much of the insolation from reaching the ground surface.



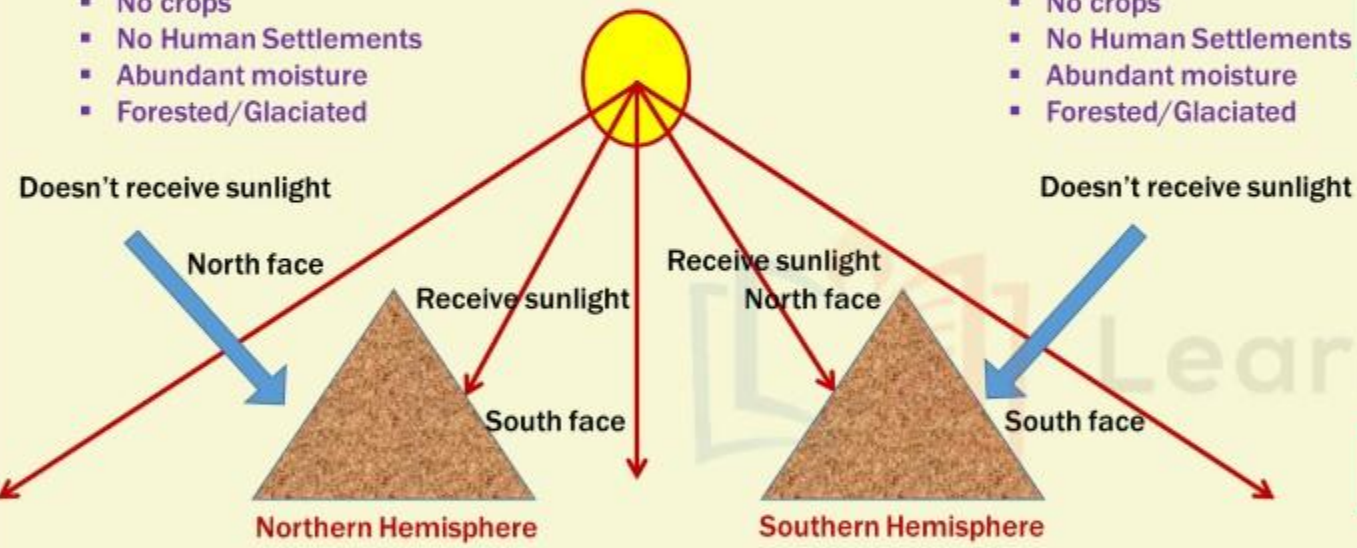
FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 4

NATURE OF THE GROUND SLOPE

- No sunlight
- No crops
- No Human Settlements
- Abundant moisture
- Forested/Glaciated



- Abundant sunlight
- Little moisture in soil due to intense heat
- Good crops
- Human Settlements

- No sunlight
- No crops
- No Human Settlements
- Abundant moisture
- Forested/Glaciated

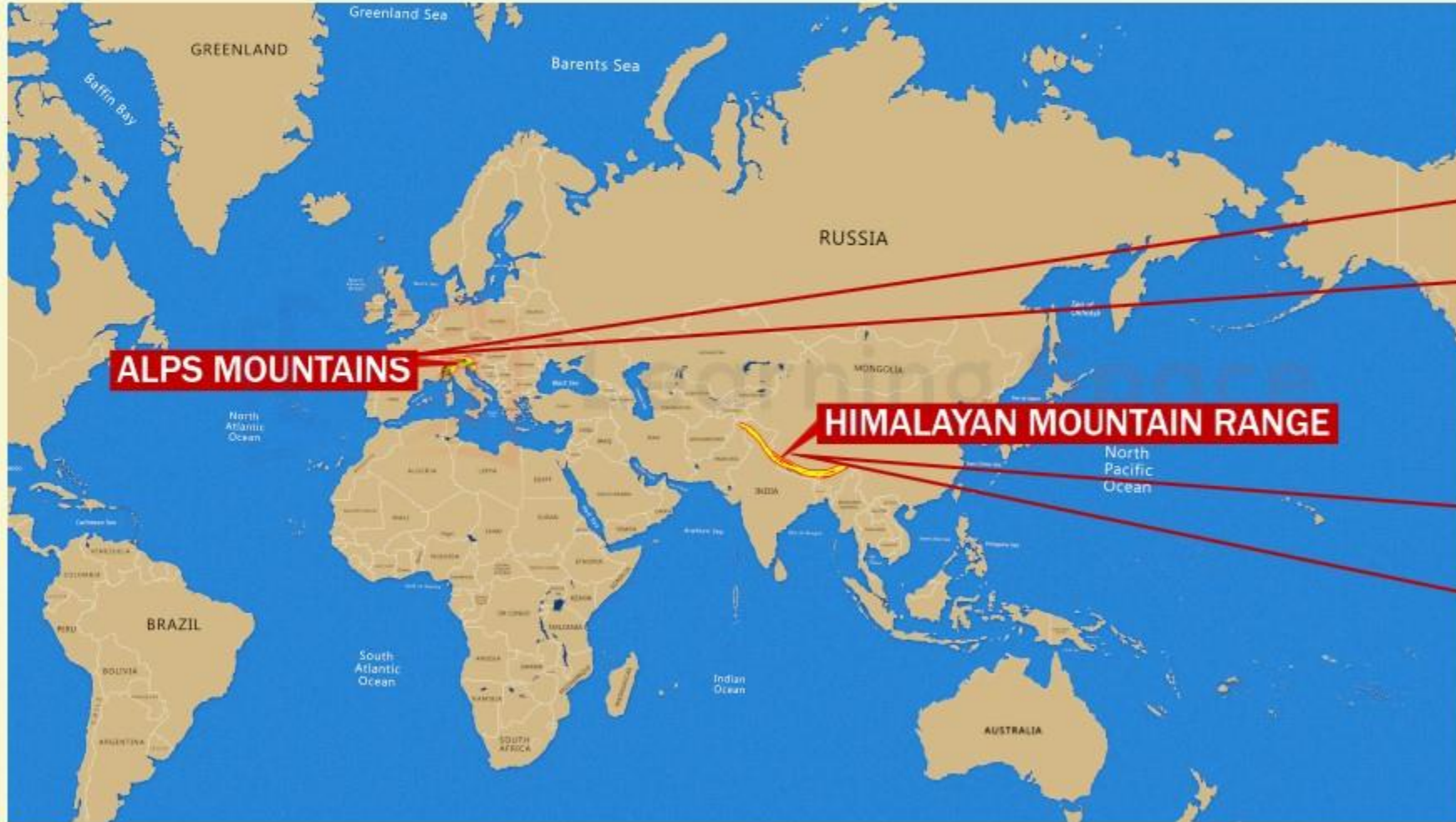
1 The ground slope facing the sun receives more insolation because the sun's rays reach the surface more or less straight and hence sun-facing ground surfaces record higher temperature than the leeward slopes where sun's rays reach more obliquely.

2 In the northern hemisphere the southward facing slopes of east-west stretching mountains receive greater amount of insolation than the northward facing slopes because of their exposure to the direct rays sun for longer duration.

Important concept for prelims/mains



FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE





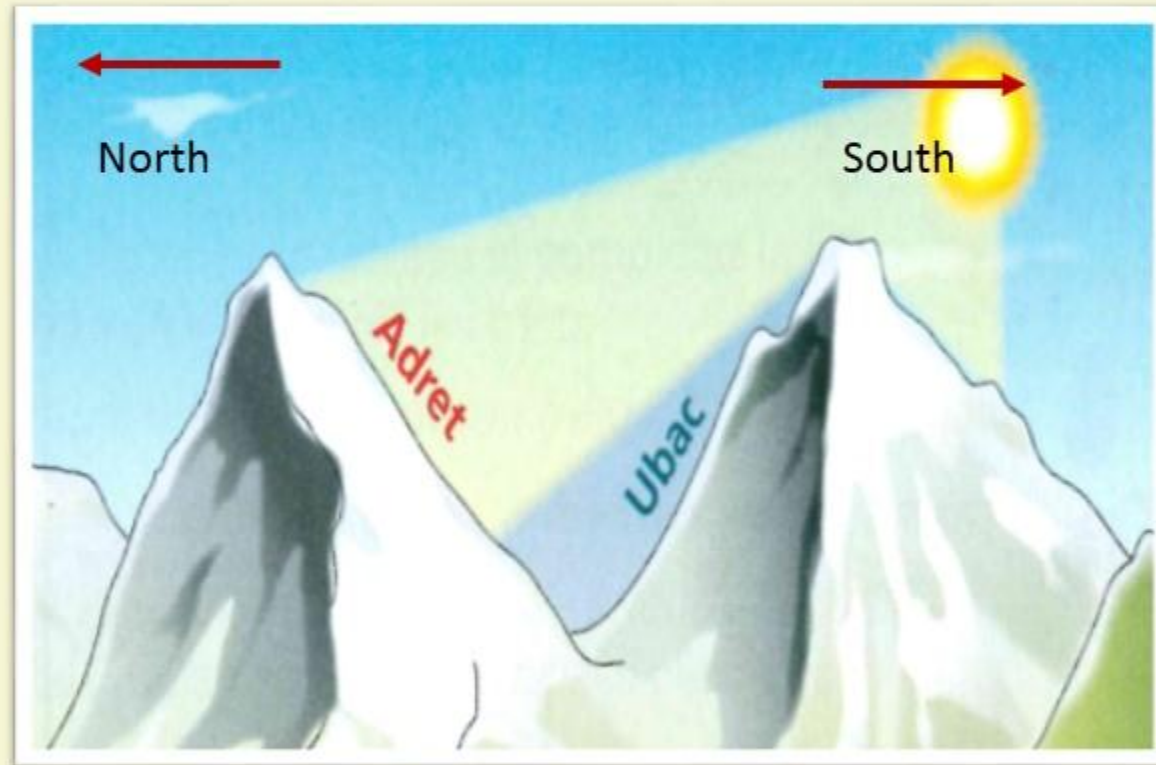
FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 4

NATURE OF THE GROUND SLOPE

“ASPECT” IN GEOGRAPHY



- 1 Aspect is the **compass direction** that a **slope faces**.
- 2 Adret is the **sunny, south-facing side** of a **mountain or mountain valley** in the **northern hemisphere**.
- 3 Adrets are **warmer, have more sun, less snow cover**. Hence it is suitable for human settlements.
- 4 The **opposite less sunny slope** is known as the **Ubac**.

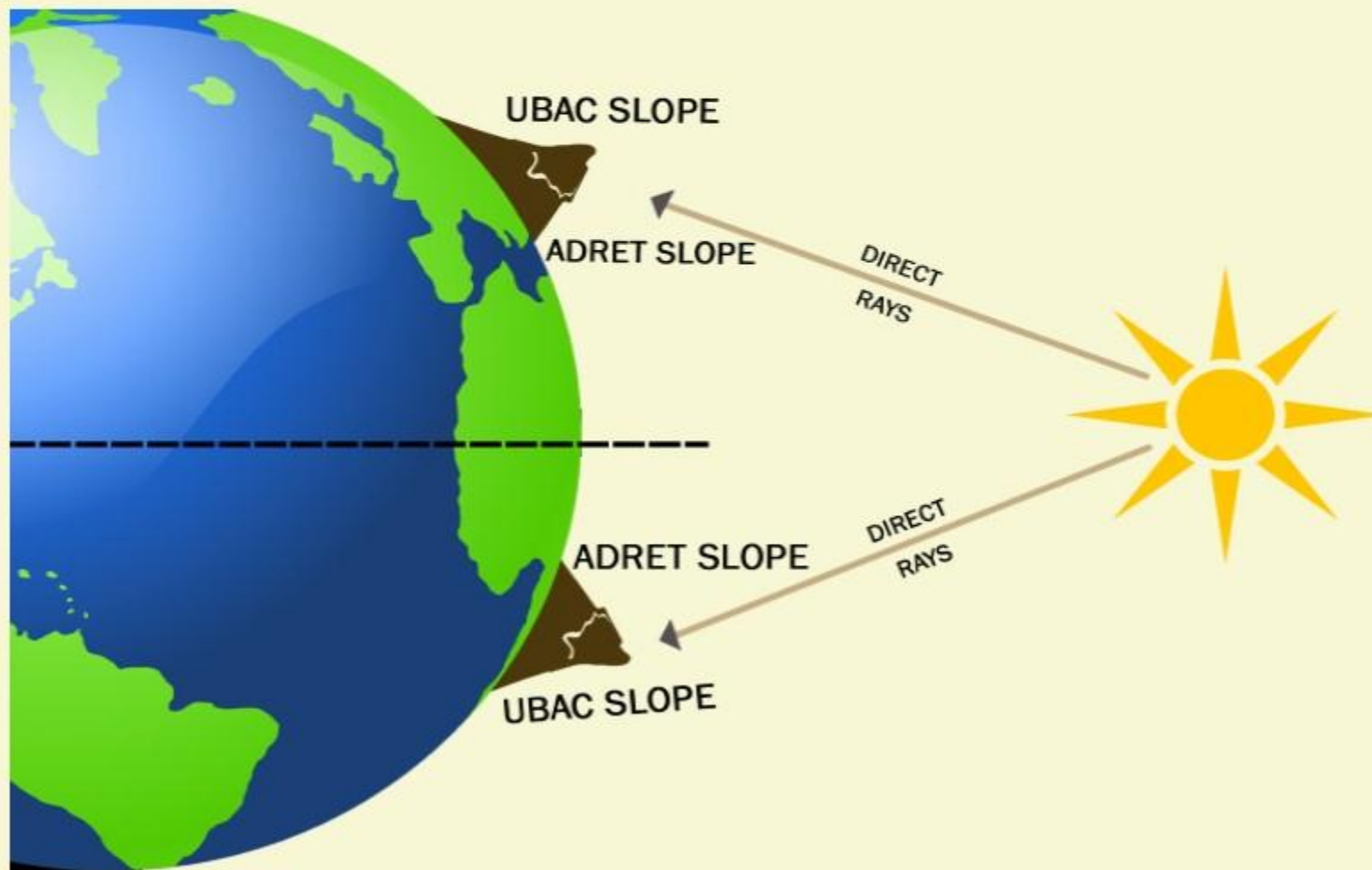


FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 4

NATURE OF THE GROUND SLOPE





FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 4

NATURE OF THE GROUND SLOPE

- No sunlight
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- No Human Settlements
- Abundant moisture
- Forested/Glaciated

Doesn't receive sunlight

North face

Receive sunlight

South face

Northern Hemisphere

- Abundant sunlight
- Little moisture in soil due to intense heat
- Good crops
- Human Settlements

- No sunlight
- No crops
- No Human Settlements
- Abundant moisture
- Forested/Glaciated

Doesn't receive sunlight

Receive sunlight

North face

South face

Southern Hemisphere

1

The ground slope facing the sun receives more insolation because the sun's rays reach the surface more or less straight and hence sun-facing ground surfaces record higher temperature than the leeward slopes where sun's rays reach more obliquely.

2

In the northern hemisphere the southward facing slopes of east-west stretching mountains receive greater amount of insolation than the northward facing slopes because of their exposure to the direct rays sun for longer duration.

Important concept for prelims/mains



FACTORS CONTROLLING THE DISTRIBUTION OF TEMPERATURE



FACTOR : 4

NATURE OF THE GROUND SLOPE



“ASPECT” IN EVERYDAY LIFE

- 1 In the Swiss Alps farming (viticulture) is much more extensive on sunny south-facing than on north-facing slopes.
- 2 In the Himalayas, this effect can be seen to an extreme degree with north-facing slopes cold, dry and much more heavily glaciated.

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suggestions@learningspace.in

To Contact us:

info@learningspace.in

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G. V. Rao

K. Srikanth

D. Sunil Kumar

L. V. Krishna

D. Chaitanya

Amrita Naidu

M. Binu Krishna

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